9a6c65ab-1047-477e-bdb6-43c3a347965f

<current>

<request>

You are working on an issue in the 'github/gh-aw-firewall' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/gh-aw-firewall/gh-aw-firewall (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-07-02T23:34:34.981+00:00</current_datetime>

Consider the following problem statement:

<problem_statement>

This section details on the original issue you should resolve

<issue_title>[Refactoring] Split assembleOptionalServices (208-line function) in src/services/optional-services.ts into per-sidecar helpers</issue_title>
<issue_description>## Refactoring Opportunity

Summary
- **File**: `src/services/optional-services.ts`
- **Current size**: 250 lines
- **Function**: `assembleOptionalServices` — 208 lines (lines 42–250)
- **Responsibilities identified**: 7 distinct concerns in a single exported function
- **Score**: 7 (mixed responsibilities +3, security-critical +2, function > 80 lines +2)

Evidence

`assembleOptionalServices` handles seven independent sidecar-

assembly blocks, all guarded by feature flags:

Lines	Block	Concern
64–83	Environment pre-sets	Injects AWF_API_PROXY_IP, AWF_CLI_PROXY_IP, AWF_NETWORK_ISOLATION
87–155	Sysroot init container	Volume filtering + sysroot-stage service + tool-cache warning (~68 lines)
162–178	iptables-init service	buildIptablesInitService + depends_on wiring
182–199	API proxy sidecar	buildApiProxyService + agent env mutations + depends_on
203–212	DoH proxy sidecar	buildDohProxyService + depends_on
216–233	CLI proxy sidecar	buildCliProxyService + agent env mutations + depends_on
241–249	Named volumes	sysroot volume assembly + agentVolumes push

The sysroot block alone is ~68 lines of volume-filtering logic with no relationship to the DoH proxy or CLI proxy blocks. The function was introduced as the extraction target of github/gh-aw-firewall#5727 but has grown to 208 lines.

Security relevance: the iptables-init and api-proxy sidecar assembly paths are security-critical (iptables NAT rules and credential-injection configuration).

Proposed Split

- Keep `assembleOptionalServices` as a thin coordinator (~40 lines)
- Extract `presetSidecarIpEnvVars(environment, config, networkConfig)` — pre-sets AWF_*_IP env vars before init container construction
- Extract `assembleSysrootService(params, agentVolumes)` — volume-filtering and sysroot-stage service wiring (~70 lines)
- Extract per-sidecar helpers for iptables-init, api-proxy, doh-proxy, and cli-proxy assembly

Affected Callers

```bash
grep -rn "from.*optional-services" src/ 2>/dev/null
```

Output: only `src/compose-generator.ts` (line 11) imports `assembleOptionalServices`. Internal helpers remain un-exported.

Effort Estimate
Medium — the sysroot block has side effects (mutates `agentVolumes` in-place via `.length = 0` + push) that require careful extraction.

Benefits
- Sysroot-stage and iptables-init assembly paths can be reviewed and tested independently
- New optional sidecars can be added without growing the monolith
- Per-block unit tests become practical

Detected by Refactoring Scanner workflow. Run date: 2026-07-02

> [!WARNING]
>

> Generated by Refactoring Opportunity Scanner · 306.4 AIC · ⊞ 6.7K · ◷
> - [x] expires on Aug 31, 2026, 4:02 PM UTC

</issue_description>
- Closes github/gh-aw-firewall#5820

<agent_instructions>create a pr that addresses this issue</agent_instructions>

Comments on the Issue (you are @copilot in this section)

<comments>
</comments>
</problem_statement>

```
Implement the necessary changes to the repository so that the
requirements specified in the <problem_statement> are met.
```